

Internship Report On
TRENDS IN HORTICULTURE CROPS IN INDIA:
with Special reference to spices and floriculture

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<u>TABLE OF CONTENT</u>		
S.NO.	DETAILS	PAGE NO.
1	ACKNOWLEDGEMENT	
2	ABOUT THE ORGANIZATION	
3	WHAT I HAVE LEARNT	
	PROJECT REPORT	
4	INTRODUCTION	1-2
5	REVIEW OF LITERATURE	2-4
6	OBJECTIVE	4
7	RESEARCH METHODOLOGY	5
8	RESULT AND DISCUSSION	5-15
9	CONCLUSION	15-16
10	SUGGESTION	16-18
11	REFERENCES	18
12	ANNEXURE	18-24

LIST OF TABLES

S. NO.	TITLE	PAGE NO.
Table 1	This table shows that the total area, production and Yield of Spices over period between 2011-12 to 2020-21 (10 years).	5
Table 2	This table shows that the total area, production and Yield of Flowers over period between 2011-12 to 2020-21 (10 years).	10
Table 1.1	Production of top produced spices in India reference with figure 1.1	19
Table 1.2	Area under top produced spices in India reference with figure 1.2	19
Table 1.3	Yield of top produced spices in India reference with figure 1.3	19
Table 1.4	Production of least produced spices in India reference with figure 1.4	20
Table 1.5	Area under least produced spices in India reference with figure 1.5	20
Table 1.6	Yield of least produced spices in India reference with figure 1.6	20
Table 1.7	Production of middle range produced spices in India reference with figure 1.7	21
Table 1.8	Area under middle range produced spices in India reference with figure 1.8	21
Table 1.9	Yield of middle range produced spices in India reference with figure 1.9	21
Table 2.1	Production of top produced Flowers in India reference with figure 2.2	22
Table 2.2	Area under top produced Flowers in India reference with figure 2.2	22
Table 2.3	Yield of top produced Flowers in India reference with figure 2.3	22
Table 2.4	Production of least produced Flowers in India reference with figure 2.4	23
Table 2.5	Area under least produced Flowers in India reference with figure 2.5	23
Table 2.6	Yield of least produced Flowers in India reference with figure 2.6	23
Table 2.7	Production of middle range produced Flowers in India reference with figure 2.7	24

Table 2.8	Area under middle range produced Flowers in India reference with figure 2.8	24
Table 2.9	Yield of least produced Flowers in India reference with figure 2.9	24

LIST OF FIGURES

S.NO.	TITLE	PAGE NO.
Figure 1.1	Production of top produced spices in India	6
Figure 1.2	Area under top produced spices in India	6
Figure 1.3	Yield of top produced spices in India	7
Figure 1.4	Production of least produced spices in India	7
Figure 1.5	Area under least produced spices in India	8
Figure 1.6	Yield of least produced spices in India	8
Figure 1.7	Production of middle range produced spices in India	9
Figure 1.8	Area under middle range produced spices in India	9
Figure 1.9	Yield of middle range produced spices in India	10
Figure 2.1	Production of top produced Flowers in India	11
Figure 2.2	Area under top produced Flowers in India	11
Figure 2.3	Yield of top produced Flowers in India	12
Figure 2.4	Production of least produced Flowers in India	12
Figure 2.5	Area under least produced Flowers in India	13
Figure 2.6	Yield of least produced Flowers in India	13
Figure 2.7	Production of middle range produced Flowers in India	14
Figure 2.8	Area under middle range produced Flowers in India	14
Figure 2.9	Yield of least produced Flowers in India	15

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I would also like to acknowledge the various researchers, academics, and institutions whose publications and other works served as important sources and references for this research. Their commitment to the study of employment and labour economics has played a crucial role in shaping the discourse on this topic.

I am grateful to **Dr. Masood Siddiqui**, Head of Department- Statistics and University of Lucknow for giving me this opportunity and encouraging me for this internship.

I would like to express my gratitude to my co-interns, friends and family for their unwavering support and encouragement throughout this research journey.

Student's Name

ARTI SINGH

2. ABOUT THE ORGANIZATION

Giri Institute of Development Studies (GIDS) is a premier centre for advanced research and training focusing on the development concerns of welcome India and its regions. Established in 1973 by a group of eminent academicians and leading public personalities and named after the former President and eminent labour leader Late Shri V.V. Giri, the Institute has emerged a vibrant centre of policy oriented and analytical multidisciplinary research in the country. GIDS is supported by Indian Council of Social Science Research (ICSSR) and Government of Uttar Pradesh as an autonomous institute of national importance.

History

The Institute, named after the former President and eminent labour leader late Shri V.V. Giri, was established by a group of eminent social scientists and leading public personalities. Late Professor V.B. Singh, a well-known labour economist was the founder Director of the Institute. In 1977 Professor T.S. Papola joined the Institute as first full time Director of the Institute. T.S. Papola joined the Institute as first full time Director and nurtured the Institute in its initial years and developed it into a well- known centre of research on development issues.

The first Board of Governors of GIDS was headed by Shri D. P. Dhar and had eminent social personalities and scholars like Naval H. Tata, Malcolm S. Adiseshiah, Baljit Singh and Nagendra Singh as its members. A wide range of distinguished intellectuals and policy makers have been associated with the governing body of the Institute. They included Professors P.N. Haksar, Yashpal, Nityanand, S.R. Hashim, Yogendra Singh, Y.K. Alagh, G.S. Bhalla, Ravi Srivastava, and K.P. Kannan among others.

Focus of Research

GIDS research covers a wide range of issues, broadly categorised into the following themes:

- Agriculture and rural development
- Labour, Employment, Livelihoods and Development
- Inter-regional and inter-state disparities.
- Population, Public Health, Education, basic amenities, and Human Development
- Poverty, Vulnerability and Social Security
- Social infrastructure particularly health, education, and basic amenities
- Rural industries and informal sector
- Urbanisation and rural-urban migration
- Scheduled Castes, Minorities, and other Socially Excluded Groups
- Gender and Child studies
- Evaluation of government schemes and programmes

FACULTY OF GIDS

Director

Professor Sanjay Singh, is the Acting director of Giri Institute of Development Studies. He has completed four major projects on Democratic Decentralization and Gender, Legal Impact Assessment and Environment and Social Impact Assessment contributing significantly to academic and public discourse. His published works include four books and more than thirty reviews and articles. He also served as the Convener of the Research Committee (RC-07) on Adivasi and Tribal Studies of the Indian Sociological Society.

Professor Parmod Kumar, was the former Director of Giri Institute of Development Studies. Previously he worked as In-charge Director and Professor, Agricultural Development and Rural Transformation Centre, Institute for Social and Economic change (ISEC), Bengaluru, India. He has published around 70 research articles in referred national and international journals. He has authored/co-authored more than twelve research volumes. Its faculty consists of four Associate professors, five Assistant professors, four library staff, one Research supporting staff, 6 Administrative staff and seven Subordinate staff.

ABOUT THE LIBRARY

The GIDS library is an invaluable resource centre for students, researchers, and faculty of Social Sciences disciplines. It is a State-of-the-Art Library which provides access to rare Academic Resources and Infrastructure for its scholars. It is rich in resources and is a well ensconced special library aiming in generating fresh ideas from learning resources in form of collections that may be textual or digital. The basic aim is to be profound in generating elevated values, enriching academic excellence, and aiding in high quality research work that matches global benchmarks. The commitment of the library is towards fulfilling its mission of facilitating access to current, global and relevant information by identifying, acquiring, organizing and retrieving information to serve the academics needs of research fraternity.

The GIDS library works with a goal to be committed to provide accurate, fast and need based user services keeping in mind the institutional objectives and ever-changing research needs of the researchers. At the same time, the quality assurance is kept in mind to render these services.

Objectives

- To create a support system for researchers
- To act as the centre of learning resources
- To strengthen the academic infrastructure of the institute
- To support the research and consultancy programmes of the institute
- To store, retrieve and disseminate information
- To have repository centre of information on the state of Uttar Pradesh
- To provide modern state- of- the-art, integrated and networked library services.

LIBRARY RESOURCES

The library has built specialized and robust collection of rare documents in social sciences catering to the research objectives of the Institute. It is an asset for

academicians and policy makers in this part of the country. The books in print form still are the prime need of the users. There is also focused subscription of on-line resources in the library which are highly used by the scholars.

Books and Journals

The GIDS Library has a specialized collection of resources keeping view the changing research and academic needs of its scholars. The library houses a total collection of 42,768 and subscribes to 90 Indian Journals apart from foreign journals subscribed through JSTOR. The collection in terms of various heads is given below:

- Major learning resources-42768
- Books-38690
- Government reports-1797
- Monographs-2946
- World Bank-8337
- Pamphlets-271
- Bound journals-7266
- Reference volumes-2490
- Ilo-225
- OCED-500
- United Nations-375
- Cmie-456
- Periodicals- List of journals

Special Collection

Microforms of GIDS publication, Institutional publications, Digitised & microforms of Gids publication, Official publications on Uttar Pradesh, World Bank depository library, Electronic Journals, Conference Alert , Contributions to Indian Sociology, Down to Earth, Economic & Political weekly, History & sociology of South Asia, India Quarterly, Indian Economic & Social History Review, Indian Journal of gender Studies,

International Journal of Rural Management, Journal of South Asian Development, Kurukshetra, Laghu Udyog Samachar, Manushi, Margin, Panchayati Raj Update, RBI bulletin, Social change, Sarvekshna and yojana. There are also variety of learning resources available like E- Resources Facilities, training/academic programs conducted by the institute.

DR. NOMITA P. KUMAR ASSITANT PROFFESOR (SENIOR SCALE) -

ECONOMICS holds master's degree in economics from Lucknow University and a PhD in economics from gids, Lucknow, Chhatrapati Sahu ji Maharaj University, Kanpur. She was awarded ICSSR doctoral fellowship at gids from 1989-1991 for her doctorate and UGC fellowship for post-doctoral research during 2003-2006. She has almost 34 years of research/ teaching experience and has worked in different research projects at the institute for state government, NABARD and UNICEF etc. She worked for second state finance commission as senior research fellow under government of Uttar Pradesh. Her research interest includes employment and labour market, gender issues, small scale industry, regional disparity and social issues related to women and girls. Many of her sole-authored research papers have been published by national/ international publishers of repute including economic and political weekly, Indian journal of labour economics, Indian journal of human development, Indian journal of regional science, Asian economic review, productivity, man and development and Madhya Pradesh journal of social science research. She has also contributed three books entitled "women entrepreneurship in urban sector", "child marriage and its consequences" and "Indian widows in search of god" and numerous research chapters to edited books viz; "regional disparities in economic and social development", in growth, disparities and inclusive development in India perspective from the Indian state of Uttar Pradesh, edited by prof R. P. Mamgai, Springer Singapore, Tracking public outlays on health care with focus on outcomes in Uttar Pradesh in health gender and development: multi- disciplinary perspectives, eds. S. Siva Raju and T.V. Sekhar: B.R. publishing corporation, Delhi and IASSH, 2016, "outlays and outcomes of ICDS as

national flagship programme in Uttar Pradesh” in national flagship programmes: issues and challenges academic foundation, new Delhi and “policy reforms and small enterprises: entrepreneur perceptions from Uttar Pradesh” in micro and small enterprises in India: the era of reforms edited by prof. Keshab Das, Routledge India, New Delhi. She has also critically reviewed as reviewer for PLOS and PONE publishers

3. WHAT I HAVE LEARNT

I had the privilege of participating in an internship program for 45 days, which provided me with a wealth of learning opportunities. The most significant lessons I discovered while serving as an intern are outlined in this report, which also discusses various aspects of my professional development. I have gained practical knowledge during my internship, as well as skills in problem-solving, adaptability, professional etiquette, time management, and prioritization. Additionally, this internship taught me how to communicate my ideas clearly and how to analyse and review other people's work.

I had the chance to learn new things and broaden my understanding. I learned how to prepare research during the research project and how to analyse the sample that was obtained from various sources. In this project, I also learned how to use data from government reports, sort it in accordance with our requirements, and conduct an analysis to reach a conclusion.

Working as an intern at GIDS was a great experience. During these 45 days internship at GIDS, we have learned a lot of new things. It enabled me to develop my professional skills. I'm grateful for the chance this internship has given me, and I am eager to put what I've learned to use in the future.

Trends in Horticulture Crops in India with Special reference to Spices and Floriculture

Introduction:

Horticulture is an important component of agriculture having significant role in the economy of the country. India's varied agro-climatic conditions provides an additional advantage in favour of growing of a wide variety of horticultural crops such as fruits & vegetables, tuber crops, plantation crops, flowers, spices & condiments etc. Commercial importance of horticulture crops has also been increasing and gaining grounds all over the world as these crops contribute significantly to the country's economy. Various horticulture crops also play an important role in human nutrition, preventing diseases and contributing to the nation's development and prosperity. In particular, the horticulture crops such as fruits and vegetables are rich source of vitamins, minerals, proteins and carbohydrates that are essential in human diet. Similarly, other crops like flowers and ornamental crops enhance aesthetic value of our environment while medicinal crops yield pharmaceutical constituents. Thus, horticulture assumes a great importance in food and nutritional security, general health and well-being of our population. Horticulture crops form a vital part in the Indian agricultural production. India is the second largest producer of fruits and vegetables in the world. Cultivation, marketing and processing of these crops generate significant employment and livelihood opportunities in many part of the country.

India is the second largest producer of fruits as well as vegetables after China. The country ranks first in the world in the production of mango, banana, sapota and acid lime. Over the years, the country has achieved highest Yield in grapes as well. Still, the country has huge untapped potential to improve upon the Yield and production of fruits, vegetables and flowers with the help of already available technological advancement in the field of agriculture. The importance of the expansion of horticulture has been increased substantially owing to its vast export potential in the WTO regime.

The term Horticulture is derived from the Latin words: “hortus” meaning garden and “cultura” meaning cultivation. In ancient days the gardens had protected enclosures with high walls or similar structures surrounding the houses. The enclosed places were used to grow fruit, vegetables, flowers and ornamental plants. Therefore, in original sense “Horticulture refers to cultivation of garden plants within protected enclosures”. At present the horticulture may be defined as the science and technique of production, processing and merchandizing of fruits, vegetables, flowers, spices, plantations, medicinal and aromatic plants. Horticulture is a wide field and includes a great variety and diversity of crops. The science of horticulture can be divided into several branches depending upon the crops it deals with.

“In this report main focus on the Trend of Area, Production and Yield of Spices and Floriculture Crops in India.”

Spices: They constitute an important group of horticulture crops and are defined as vegetable products or mixture thereof, free from extraneous matter used for flavouring, seasoning and imparting aroma in foods. India is known as home of spices producing a wide variety of spices like black pepper, cardamom, ginger, turmeric, chilli, Coriander etc. Major spice producing states are Kerala, Andhra Pradesh, Gujarat, Rajasthan, Maharashtra, Karnataka, Orissa, etc.

They are classified as vegetable products or mixtures of vegetable products, devoid of extraneous materials, utilized for flavoring, seasoning, and scent imparting in meals, and they make up a significant section of horticultural crops. India is renowned for being the world's largest producer of spices, including coriander, ginger, turmeric, black pepper, and cardamom.

Floriculture: Flower cultivation is being practiced in India since ages. It is an important/integral part of socio-cultural and religious life of Indian people. It has taken a shape of industry in recent years. India is known for growing traditional flowers such as jasmine, marigold, chrysanthemum, tuberose, crossandra, aster, etc. Commercial cultivation of cut flowers like, rose, orchids, gladiolus, carnation, anthurium, gerbera is also being done. The important flower growing states are Tamil Nadu, Karnataka, Andhra Pradesh, Maharashtra, West Bengal, Sikkim, Jammu & Kashmir, and Meghalaya etc.

India has always engaged in the practice of flower growing. It is an essential component of Indian people's sociocultural and religious lives. In recent years, it has taken on the form of an industry. Traditional flowers like jasmine, marigold, chrysanthemum, tuberose, crossandra, aster, etc. are grown in India. Cut flowers including roses, orchids, gladioli, carnations, anthuriums, and gerberas are also grown for commercial purposes.

1. Benchmark survey on Horticulture crops by Nomita P. Kumar and Kavita Baliyan.
2. Horticulture schemes and Benefits of Farmers: IJCRT ISSN 2320-2882

Review of Literature:

Many studies have focused on the Horticulture crops in India. Some of the studies reference with the trend in area and production of horticulture in India.

Kondal (2014) in this study, has analysed inconstancy of all horticulture crops in India and to check the relationship between horticulture sector's area, production and

horticulture sector export from India and import to India. The study suggests that high and strong relationship between horticulture production and horticulture export from India. Kondal says that coefficient of variation value of mushrooms production is very low compared to other horticulture crops. Study observed that horticulture exports has increased from 5184.21 crores in the year 2001- 2002 to 17371.19 crores in the year 2010-2011 and imports are also increased due to decrease in the Yield of horticulture crops. The adaptation of modern technology in horticulture crops Yield has increased.

Sidhu (2005) this study, ‘fruit and vegetable processing industry in India: an appraisal of the post – reform period’ has examined fruit and vegetable product processing industry in India. The study period has 1991 to 2001. Data have taken various sources like annual report of the department of food processing industries, ministry of agriculture, economic survey and statistical outline of India. Study said that production of processed fruits and vegetables have been increased which was in 1991 2.80 lakh tones to increase in 2001 9.90 lakh tones. Study find out that number of licensed processing units under the fruits product order have increased in the post- reform period and these all units have been positive affected of the fruits and vegetables industry. Study said that export of processed fruits and vegetables have also increased in the study period due to globalization and liberalization. Major exported destination of fruits and vegetables product to Saudi Arabia, United States of America, Netherlands, Kuwait, Germany, United Kingdom, Egypt, Sri Lanka, United Arab Emirates, Turkey, Spain, Indonesia, Russia and France. Further the study said that these products purchased by the high class people not poor or lower class people due to value addition and high costs. But the suggested that after postreform fruits and vegetables industries have increased and APEDA has also dominant role in the export of these products.

Roy and Sengupta (2011) in this study, ‘behaviour of India’s horticultural exports: does price competitiveness play a determining role?’ has analysed the export of horticultural products from India during 1961 to 2005. In this study price data have taken in FAOSTAT. Augmented dickey-fuller, time series and co integration test have been used. The study suggested that export of horticulture goods depend on production, export price, world unit price, and world demand and producer price. Thus export of horticulture depends on their price factor and some other factor like Yield, transportation cost, packaging and his quality. Further the study find out that most of horticulture export determine the relative export price in the long run and short run. Export of banana, coffee and spices have positive and significant impact on the production of horticulture in the long run.

Kondal (2014) in this study, ‘performance of horticulture sector in India’ has analysed the performance of horticulture sector in India and introduce to the effective factors on horticulture export and imports. The study ranged from 2001- 2002 to 2010 – 2011. In this study Annual Growth Rate, Regression, Coefficient of Correlation and Percentage Method have been used. Kondal advocates that export of horticulture increased in total agriculture export during 2001-2002 to 2009-2010. But export growth rate of

horticulture was weakening after 2008-2009. The study suggested there is a good significant relationship between export/import of horticulture and agriculture Export/Import and total national Export/Import in India. Study talks about there are a good and significant effect of production and Exchange Rate on the horticulture exports and imports in India. During the study period Maharashtra has big role in the contribution of major horticulture crops in terms of area, production and Yield in India.

Choudhary (2013) in this study, 'Contribution of national horticulture mission in agriculture development' talks about 4 % targeted of horticulture sector achieve by the development of horticulture crops which have dominant role within agriculture sector. Study advised that after launching the national horticulture mission (NHM) horticulture production have been increased and generate more employment for its farming sector. Further study advocated that more foreign exchange will come by the export of horticulture production.

Aneja (2013) in this study, 'Horticulture development in India: A multi aspects Analysis' has analyzed important role of horticulture crops in India and state level role of horticulture development in India. Data have been taken from Indian horticulture board (NHB), Director of economics and statistics, Center for monitoring on Indian economy (CMIE), and export data taken from agricultural and processed food products export development authority, directorate general of foreign trade. Revealed comparative advantage and percentage method have been used for this study. The study period has taken 1991 to 2010. The study finds out that fruits and vegetables are dominant role in the horticulture crops and their area and production have been increased. Maharashtra and Andhra Pradesh have more developed than other state in the production of horticulture crops. The study suggested that grapes and mango exports earn the maximum foreign exchange for the India and onion, potato are important role in the fresh vegetables. Further the study suggested that India is not keeping revealed comparative advantage in the exports of fruits and vegetables besides Mosambi, Grapes, Mango and Onions.

Objective:

The Research aim to address the following key objectives:

1. To analyse the trend of Area, Production, and Yield of different spices and flowers of India.
2. To analyse the pattern and emerging changes in area and production of different spices and flowers over the period of time.
3. To analyse the pattern and emerging changes in yield of different spices and flowers over the period of time.
4. To suggest about the types of measures to be initiated for maximizing area and production of horticulture crops.

Research and Methodology:

In order to fulfil the above objectives, this study reference with the secondary data relating to the area of cultivation, production of different spices and flowers of different states in India. Secondary data has been compiled from different government sources that “**Department of Agriculture and cooperation and Farmers welfare**” and “**Ministry of Agriculture and Farmers welfare**”. The study covers the period between 2011-12 to 2020-21 i.e., 10 years.

The Yield of spices and flowers are calculated by the given formula are,

$$\text{Yield} = \text{Total Production} / \text{Total Area}$$

Where, production in ‘000 MT unit and area in ‘000 Ha unit.

Result and Discussion:

In this section of Result and Discussion, we have to shows that area, production and Yield of those crops which increase the Yield over 10 years (2011-12 to 2020-21).

Spices: There are seventeen spices mainly grown in India that are divided in three different groups reference with the production that are top produced, least produced, and middle or moderately produced spices.

Title: This table shows that the total area, production and Yield of Spices over period between 2011-12 to 2020-21 (10 years).

Years	Total Area in '000 Ha	Total Production in '000 MT	Total Yield in 'MT/Ha
2011-12	3212.471	5951.458	1.85261
2012-13	3075.90	5739.72	1.866028
2013-14	3131.43	5905.24	1.885797
2014-15	3192.31	5610.88	1.757623
2015-16	3473.53	6988.47	2.01192
2016-17	3668.41	8118.57	2.213101
2017-18	3876.48	8497.31	2.192
2018-19	4067.03	9499.75	2.3358
2019-20	4,350.84	10,297.86	2.3669
2020-21	4456.94	11117.34	0.291064

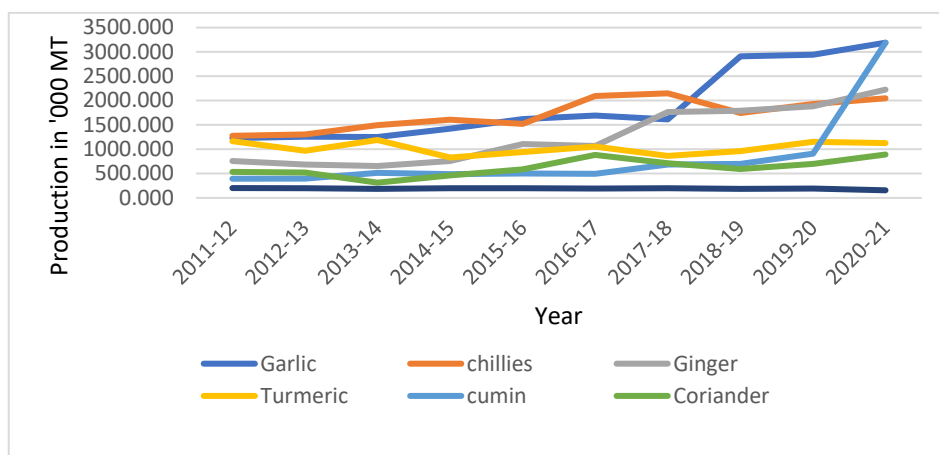
Source: Department of Agriculture and Family Welfare

Table:1

Tabel 1: Shows that the total area, production and Yield of spices. It shows that when area increases year by year then we seen that production and Yield also increases. So when we want to increase the production and Yield then we increases the area.

Top produced spices in India are Garlic, Chillies, and Ginger followed by Turmeric, Cumin, Coriander and Tamarind with the production of 19129.924, 17166.761, 12689.248, 10257.493, 8273.998, 6195.247, 1922.094 (in '000 MT) respectively under the area 3003.771, 7704.112, 1609.393, 2311.216, 8881.410, 5544.280, and 514.418 (in '000 Ha) respectively.

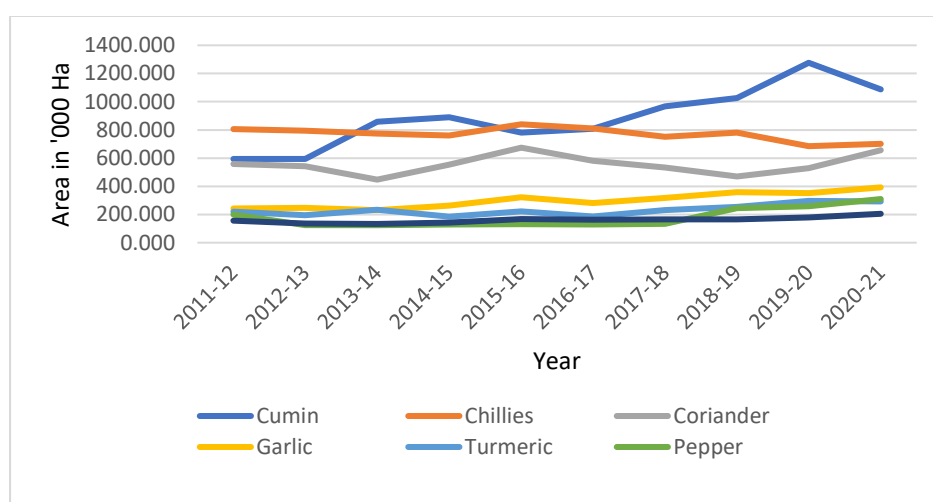
Title: Production of top produced spices in India



Source: Department of Agriculture and Family Welfares

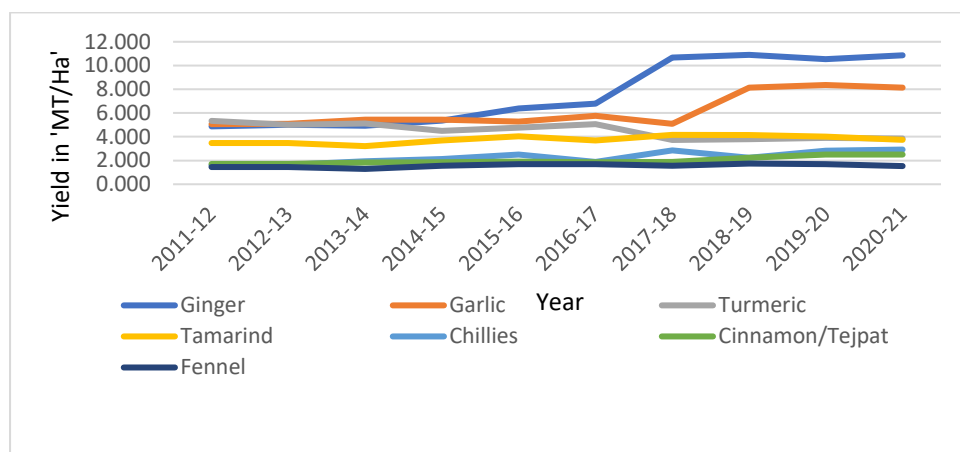
Fig 1.1

Title: Area under top produced spices in India



Source: Department of Agriculture and Family Welfares

Fig 1.2

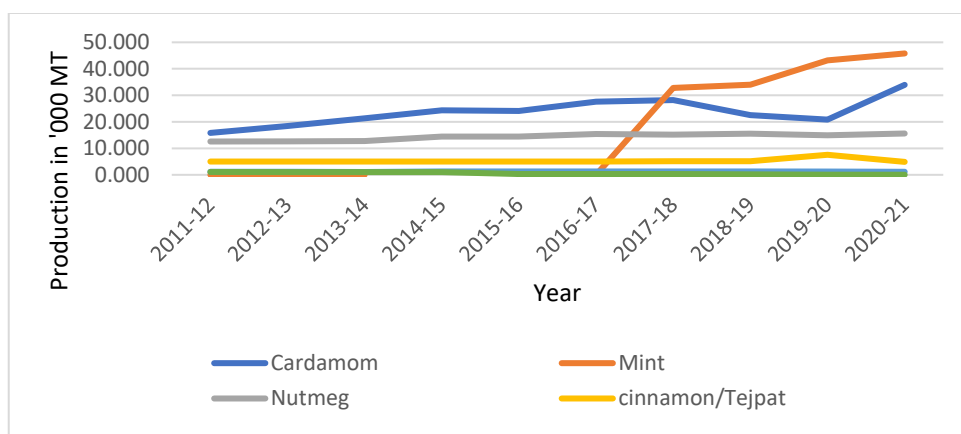
Title: Yield of top produced spices in India

Source: Department of Agriculture and Family Welfares

Fig 1.3

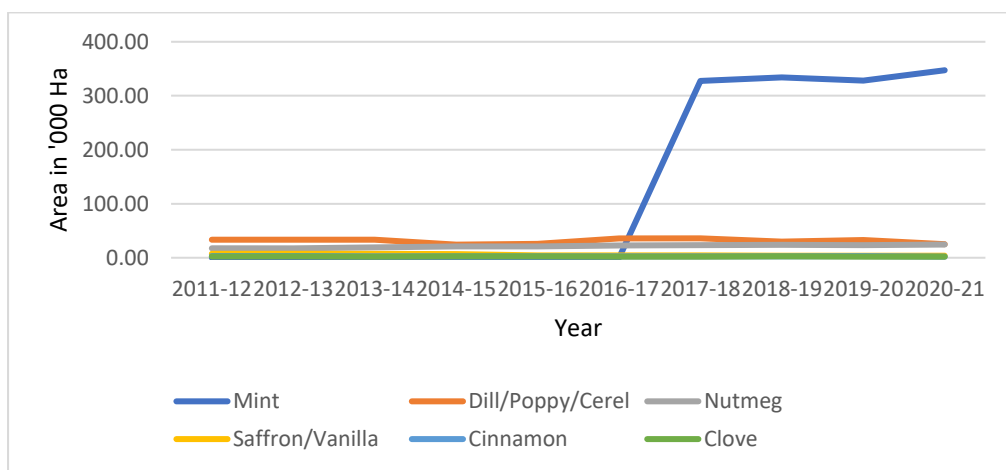
In **Fig.1.1**, **Fig.1.2** and **Fig.1.3** shows that the trend line of production, area, and Yield of top produced spices in India. In all three figures we seen that the production, area and Yield of **Ginger, garlic and chillies** are the **top produced** spices in India.

Least produced spices in India are **Cinnamon/Tejpat, Clove, Safron/Vanilla** followed by **Cardamom, Mint and Nutmeg** with production of 52.935, 11.795, 5.474, 237.276, 155.820, 143.384 respectively (in '000 MT) under the area 26.764, 22.297, 52.715, 879.876, 1336.580, 214.255 respectively (in '000Ha). Also seen that the **Mint** suddenly started being produced at year 2017-18.

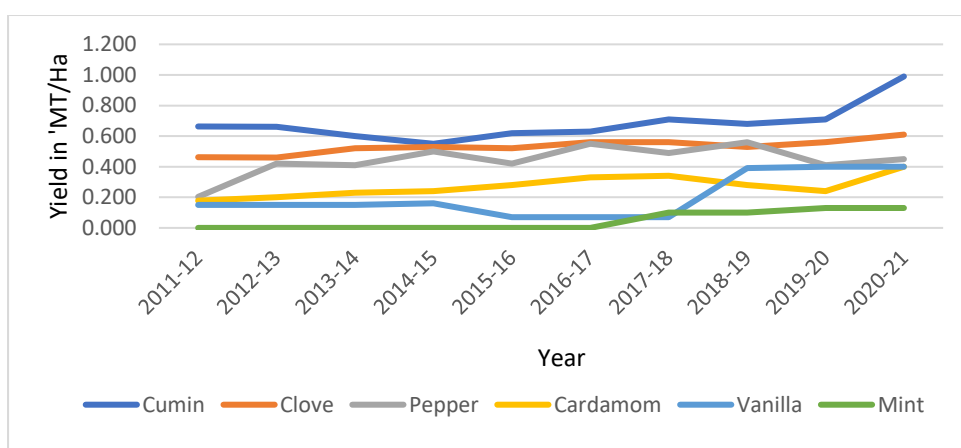
Title: Production of least produced spices in India

Source: Department of Agriculture and Family Welfares

Fig 1.4

Title: Area under least produced spices in India

Source: Department of Agriculture and Family Welfares

Fig 1.5**Title: Yield of least produced spices in India**

Source: Department of Agriculture and Family Welfares

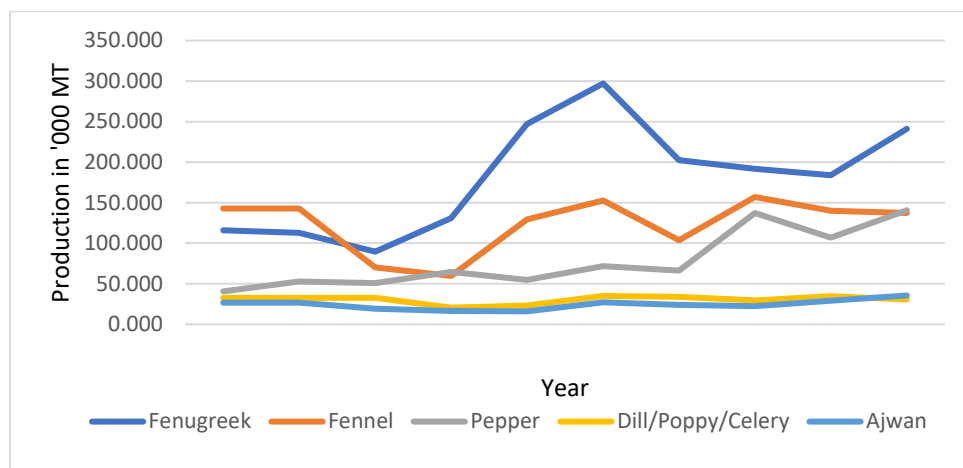
Fig 1.6

In **Fig.1.4**, **Fig.1.5** and **Fig.1.6** shows that the trend line of production, area, and Yield of least produced spices in India. In all three figures we seen that the production, area and Yield of **Cinnamon/Tejpat, Clove, Saffron/Vanilla** are the **least produced** spices in India and also seen that the **Mint** suddenly started being produced at year 2017-18.

Middel range produced spices in India are **Fenugreek, Fennel, Pepper** followed by **Dill/Poppy/Cerely**, and **Ajwan** with production of **1812.669, 1236.219, 786.302,**

305.522, 243.178 respectively (in '000MT) under the area **1359.225, 780.354, 309.660** respectively (in '000Ha).

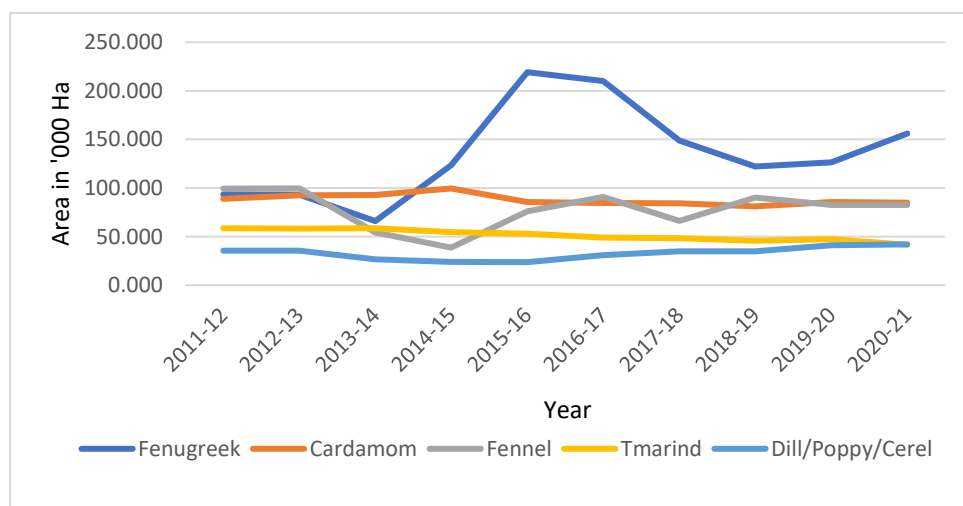
Title: Production of Middel range range produced spices in India



Source: Department of Agriculture and Family Welfares

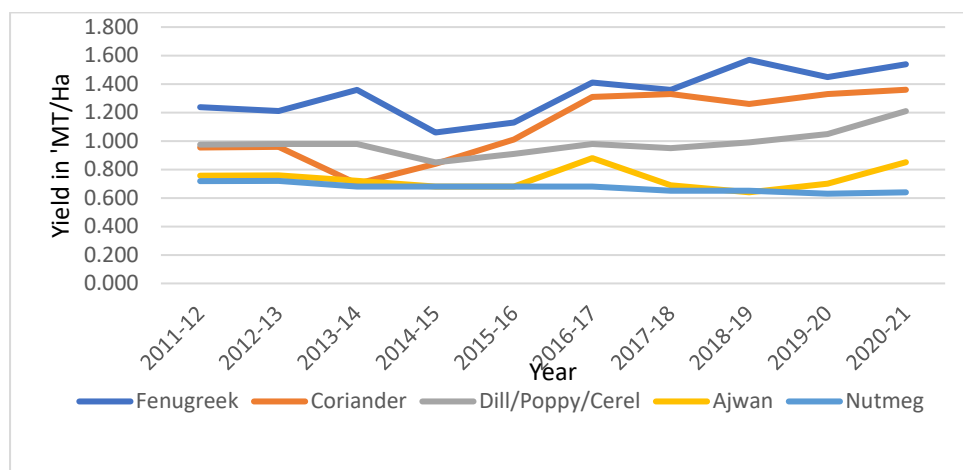
Fig 1.7

Title: Area under Middel range range produced spices in India



Source: Department of Agriculture and Family Welfares

Fig 1.8

Title: Yield of Middel range range produced spices in India

Source: Department of Agriculture and Family Welfares

Fig 1.9

In **Fig.1.7**, **Fig.1.8** and **Fig.1.9** shows that the trend line of production, area, and Yield of least produced spices in India. In all three figures we seen that the production, area and Yield of **Fenugreek, Fennel, Pepper** are the **Middel range produced** spices in India.

Flowers: There are eleven flowers mainly grown in India that are divided in three different groups reference with the production that are top produced, least produced, and middle or moderately produced flowers.

Title: This table shows that the total area, production and Yield of Flowers over period between 2011-12 to 2020-21 (10 years).

Years	Total Area in '000 Ha	Total Production in '000 MT	Total Yield in 'MT/Ha
2011-12	253.65	76717.59	302.4582
2012-13	232.74	78461.06	337.1129
2013-14	255.02	2297.026	9.007167
2014-15	246.13	1744.37	7.087174
2015-16	277.57	2183.91	7.867943
2016-17	303.31	2395.83	7.899054
2017-18	324.00	2635.91	8.135454
2018-19	303.21	2909.72	9.596449
2019-20	323.33	2999.71	9.277699
2020-21	322.02	2980.04	9.254188

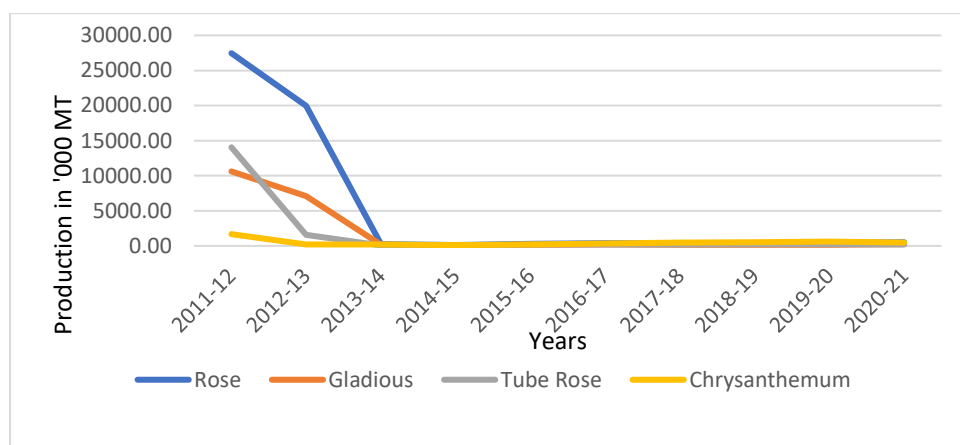
Source: Department of Agriculture and Family Welfares

Table:2

Tabel 2: Shows that the total area, production and Yield of Flowers. It shows that when area increases year by year then we seen that production and Yield also increases. So when we want to increase the production and Yield then we increases the area.

Top Produced Flowers in India are **Rose, Gladious, Tube rose, and Chrysanthemum** with the production of **50188.13, 18984.12, 16997.22 and 4774.15** respectively (in '000MT) under the area **318.54, 114.83, 150.77 and 202.41** respectively (in '000Ha).

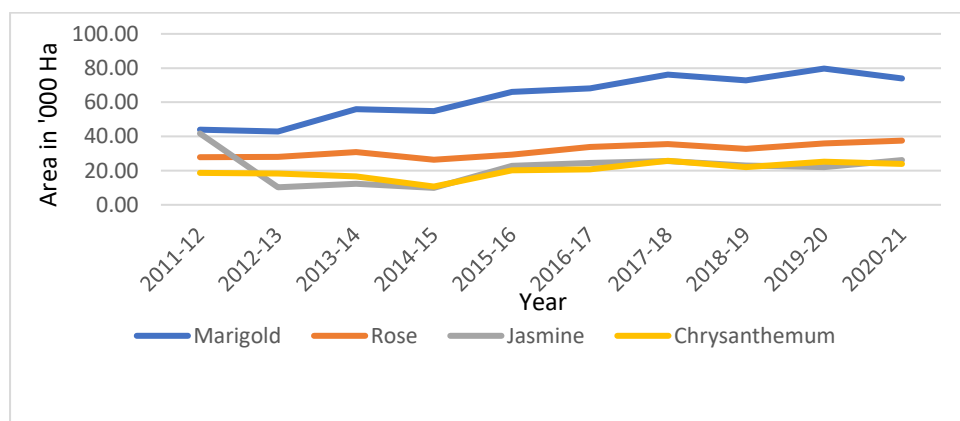
Title: Production of top produced Flowers in India



Source: Department of Agriculture and Family Welfares

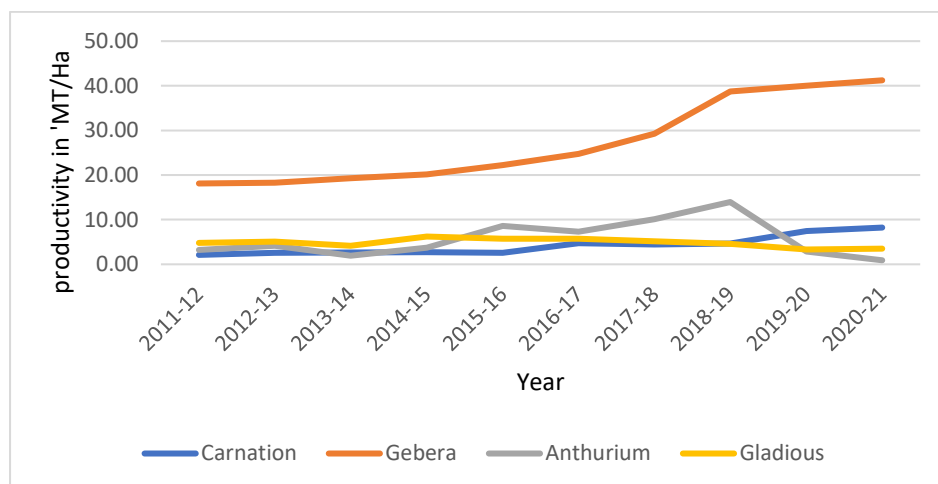
Fig 2.1

Title: Area under top produced Flowers in India



Source: Department of Agriculture and Family Welfares

Fig 2.2

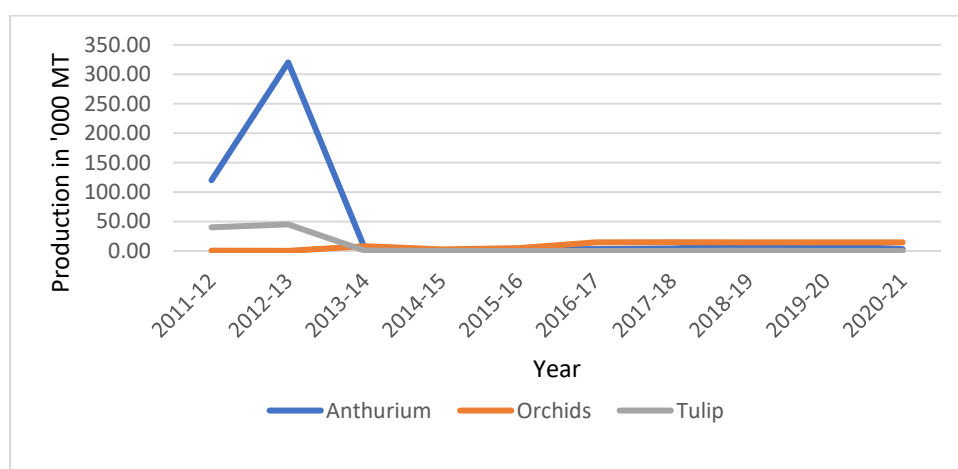
Title: Yield of Top produced Flowers in India

Source: Department of Agriculture and Family Welfares

Fig 2.3

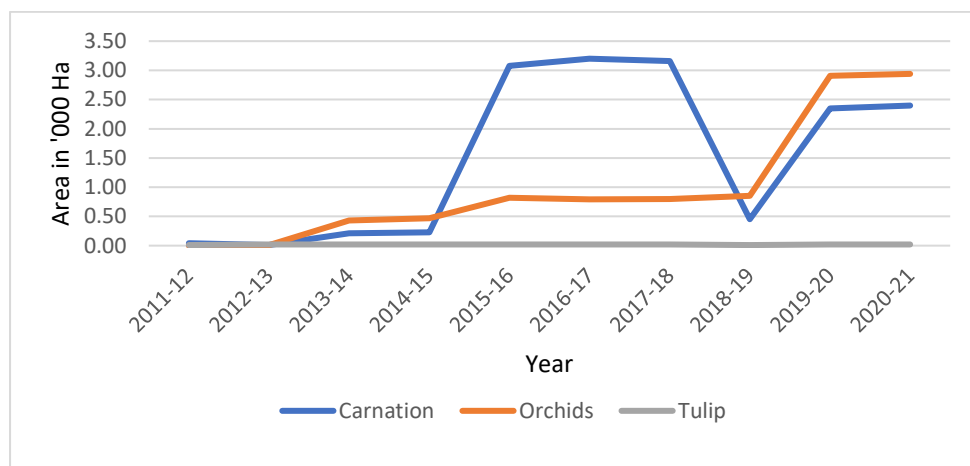
In **Fig.2.1**, **Fig.2.2** and **Fig.2.3** shows that the trend line of production, area, and Yield of Top produced Flowers in India. In all three figures we seen that the production, area and Yield of are **Rose, Gladios, Tube rose, and Chrysanthemum** the **Top produced** Flowers in India.

Least produced flowers in India are **Anthurium, Orchid, and Tulip** with the production of **476.37, 89.05, and 86.02** respectively in ('000MT) under the area of **8.35, 10.02, and 0.18** respectively (in '000Ha).

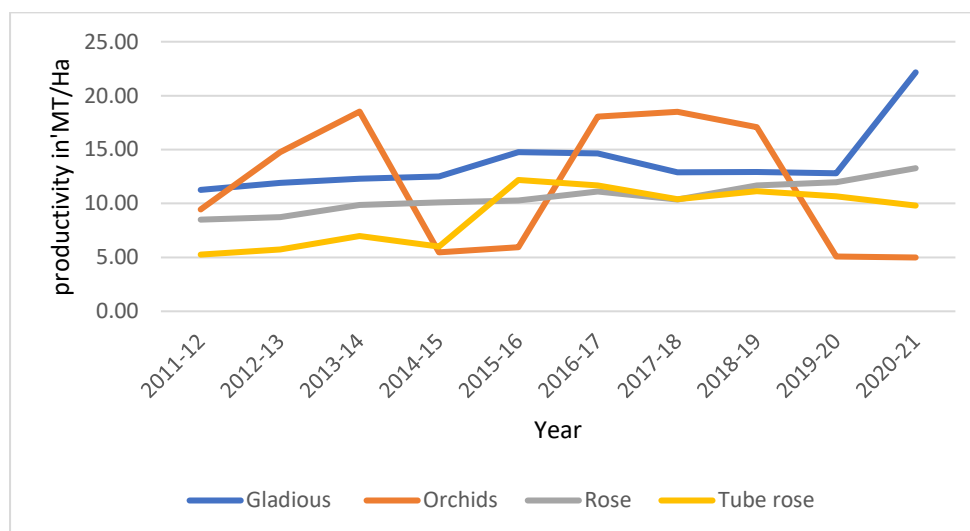
Title: Production of Least produced Flowers in India

Source: Department of Agriculture and Family Welfares

Fig 2.4

Title: Area under Least produced Flowers in India

Source: Department of Agriculture and Family Welfares

Fig 2.5**Title: Yield of Least produced Flowers in India**

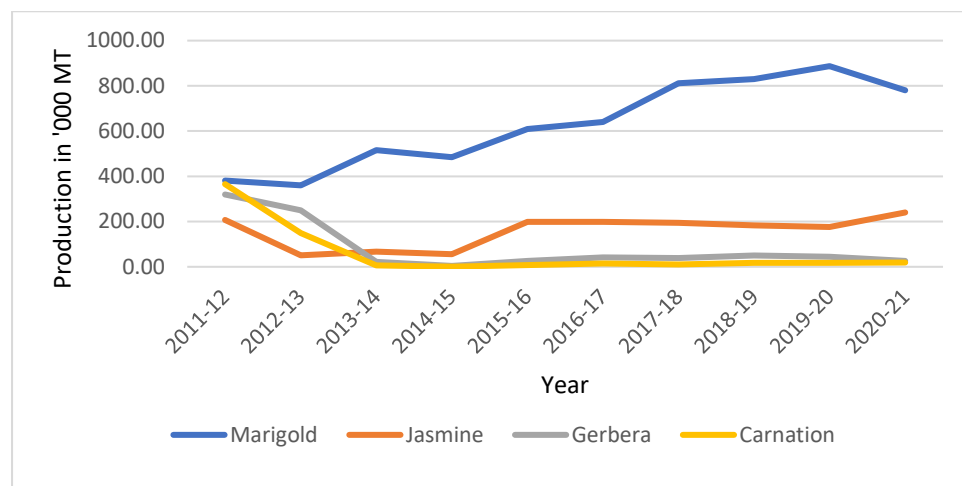
Source: Department of Agriculture and Family Welfares

Fig 2.6

In **Fig.2.4**, **Fig.2.5** and **Fig.2.6** shows that the trend line of production, area, and Yield of Top produced Flowers in India. In all three figures we seen that the production, area and Yield of are **Anthurium, Orchid, and Tulip** are the **Least produced** Flowers in India.

Middel range Produced Flowers in India are **Marigold, Jasmine, Gerbera and Carnation** with Production of **6300.59, 1573.97, 821.17, and 610.99** respectively (in '000MT) under the area of **634.31, 218.34, 18.19 and 15.13** respectively (in '000Ha).

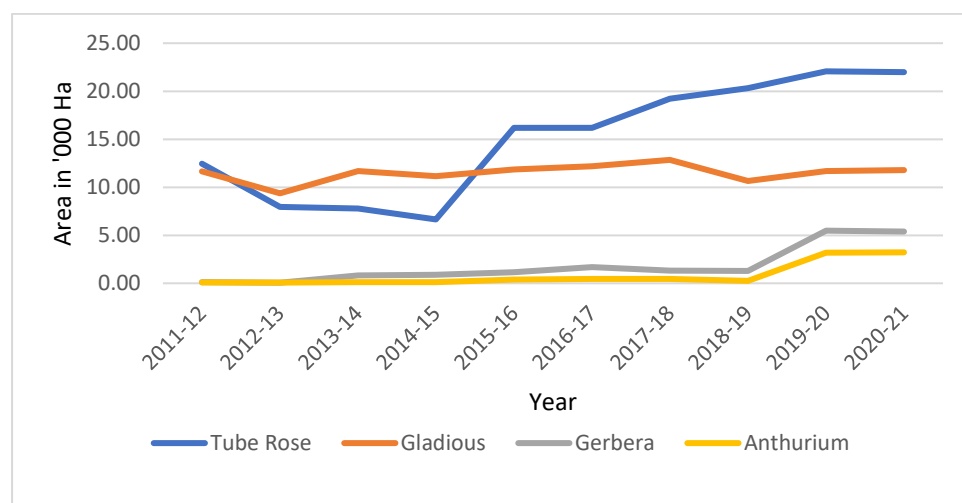
Title: Production of Middel range produced Flowers in India



Source: Department of Agriculture and Family Welfares

Fig 2.7

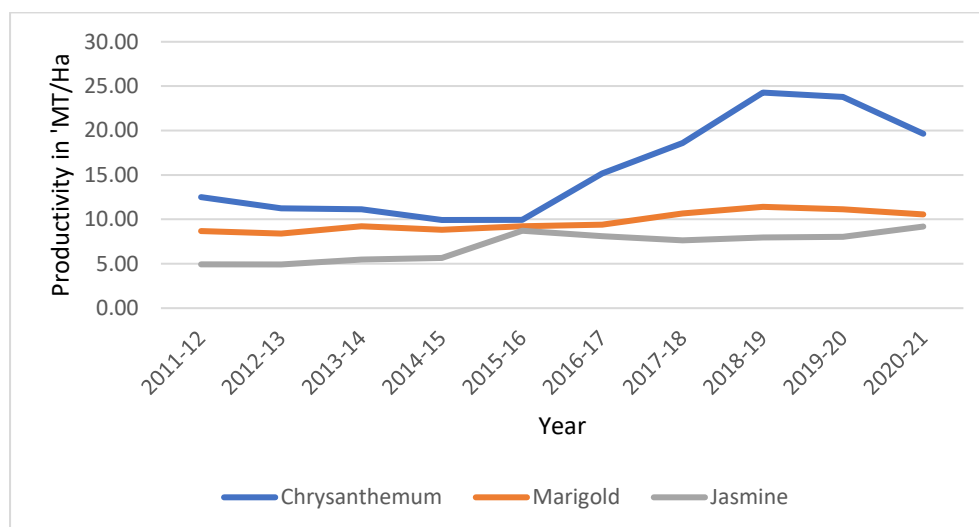
Title: Area under Middel range produced Flowers in India



Source: Department of Agriculture and Family Welfares

Fig 2.8

Title: Yield of Middel range produced Flowers in India



Source: Department of Agriculture and Family Welfares

Fig 2.9

In **Fig.2.7**, **Fig.2.8** and **Fig.2.9** shows that the trend line of production, area, and Yield of Top produced Flowers in India. In all three figures we seen that the production, area and Yield of are **Marigold, Jasmine, Gerbera and Carnation** are the **Middel range produced** Flowers in India.

Conclusion:

The horticulture sector is known worldwide and has a very vital significance in the economic development for the different State of India. We have studied the data of area and production of spices and flowers grown in India and observed that the production of various horticulture products mainly spices and Flowers are on rise. Diversification towards the spices and flowers is seen not only in production terms but also in terms of area as well. The increase in the production is not only because of the increase area under production, it is also due to the rapid increase in Yield of the spices and flowers. The growth in the Yield along with the area under production has led to an improvement in the share of horticulture commodities. They constitute an important group of horticulture crops and are defined as vegetable products or mixture thereof, free from extraneous matter used for flavouring, seasoning and imparting aroma in foods. India is known as home of spices producing a wide variety of spices like black pepper, cardamom, ginger, turmeric, chilli, Coriander etc. Major spice producing states are Kerala, Andhra Pradesh, Gujarat, Rajasthan, Maharashtra, Karnataka, Orissa, etc.

They are classified as vegetable products or mixtures of vegetable products, devoid of extraneous materials, utilized for flavoring, seasoning, and scent imparting in meals,

and they make up a significant section of horticultural crops. India is renowned for being the world's largest producer of spices, including coriander, ginger, turmeric, black pepper, and cardamom, chillies.

India is known for growing traditional flowers such as jasmine, marigold, chrysanthemum, tuberose, crossandra, aster, etc. Commercial cultivation of cut flowers like, rose, orchids, gladiolus, carnation, anthurium, gerbera is also being done. India has always engaged in the practice of flower growing. It is an essential component of Indian people's sociocultural and religious lives. In recent years, it has taken on the form of an industry. Traditional flowers like jasmine, marigold, chrysanthemum, tuberose, crossandra, aster, etc. are grown in India. Cut flowers including roses, orchids, gladioli, carnations, anthuriums, and gerberas are also grown for commercial purposes.

From the above graphs we can conclude that Yield of mostly Spices and Flowers are increasing over the ten years from 2011- 20 along with increase in area under production in some Spices and Flowers.

One of the reasons of such rapid increase in production is, as farmers are adopting the cultivation of Spices and Flowers in place of traditional crops and improving their Yield by using best quality of seeds, suitable soil composition and controlled temperature etc.

The second reason for rapid increase in the production is that cultivation of these crops generates lot of employment opportunities for the local labours in the rural areas. And they give more profit than the traditional crops.

The graphs mentioned above helped us in visualizing the trend of production, area and Yield over the time of ten years. The importance of analysing the trend and patterns in the increase or decrease in the production, area and Yield. This analysis help growers or farmers to effectively plan the further production techniques, these analyses also help farmers to choose which crop they wanted to grow to optimize the profit.

Suggestion:

The different states of India, enjoys comparative advantage in horticulture crops because of its favourable agro-climatic factors suitable for growing a large variety of horticulture crops. Horticultural output has increased at a fairly encouraging rate in the recent years due to the impetus provided by growing demand for these products. This market led process is still too small to have a large impact on the agricultural economy of the state and needs to be supported through public policy in a systematic manner.

These are the various recommendations suggested in order to improve the Yield of the crops and also to help in increasing the income of the farmers engaged in cultivation of various horticulture crops. It can be said that if efforts are made in proper and effective way, its weakness can be converted into its opportunities and also help in improving the socioeconomic condition of the growers involved in the farming. It can be concluded that if farmers shift their area to horticulture crops, income can be increased and can be more profitable to the farmers to grow various type of crops in the districts. The policy package to promote the process of crop diversification should include among other things the following:

Improvement in the rural infrastructure particularly in the field of storage, transport and marketing of these crops: It has been reported that more than 20-30 per cent of the produce is lost in the post-harvest operations. This is mainly because of factors like non-availability of proper infrastructure in terms of facilities for handling the produce, poor transport & road condition and lack of storage and ware-house facilities discourage the farmer to diversify their crop. Government would have to invest on public transport, roads and most important on cold storages and warehouses to motivate the growers to diversify their crops and increase the area under horticulture crops. Irrigation was the major problem revealed by the farmers in the districts. Hence, adequate and regular irrigation facility should be provided by the government.

Minimum Support price System for Horticulture Crops also: The horticulture growers faces the problem related to the prices of different crops. Government must interfere to decide the minimum price of all horticulture crops as the price of wheat and rice. It will give a kind of security to the growers specially vegetables growers.

Data and information: It is suggested that data on area, production and yield also need to be streamlined after conducting proper survey. Data generated by the Department and other agencies at the field level vary to a large extent. Similarly, there is a need to streamline the data and put in place a system to update and publish the data on market arrival and prices of different horticulture crop-wise.

Undoubtedly, the area, production of horticulture crops has improved to some level, but yield was not up to the mark as expected. Therefore, serious policy efforts are needed to increase the potential and improve the Yield of various crops at state and district level.

Improve the staff at State level as well as District level: Horticultural departments are sufferings from the lack of staff at the state level as well as district level for providing the required extension services. The Horticulture Department of the State Government

has been facing shortage of staff. There is inadequate awareness on post-harvest needs and its technology among the mango growers and traders.

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National Horticulture Board, Ministry of Agriculture and Farmers Welfare Government of India
<http://nhb.gov.in/Statistics.aspx?enc=WkegdyuHokljEtehnJoq0KWL79sOQCy+W4MfOk01GFOWQSEvtp9tNHHoiv3p49g>, <http://agricoop.nic.in/statistics/state-level>

Annexure:

Area, production and Yield of Spices

Titel: Production of top Produced Spices in India reference with the Fig. 1.1

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Garlic	1228.324	1259.27	1251.88	1425.46	1617.34	1693.20	1610.62	2909.90	2,944.15	3189.78	19129.924
chillies	1276.301	1304.38	1492.14	1605.01	1520.39	2095.49	2149.23	1743.31	1,931.30	2049.21	17166.761
Ginger	755.618	682.63	655.06	760.31	1109.18	1067.64	1761.57	1788.21	1,884.19	2224.84	12689.248
Turmeric	1166.843	971.09	1189.89	830.39	943.30	1055.63	862.75	960.73	1,153.01	1123.86	10257.493
cumin	394.328	394.33	513.85	485.51	503.27	492.88	688.66	699.45	911.94	3189.78	8273.998
Coriander	532.947	523.88	313.65	461.71	584.98	883.23	709.84	592.46	701.23	891.32	6195.247
Tamarind	202.574	201.82	188.13	201.66	197.15	194.41	201.27	188.92	189.87	156.29	1922.094

Source: Department of Agriculture and Family Welfare

Table: 1.1**Titel: Area under Top Produced Spices in India reference with the Fig. 1.2**

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Cumin	593.980	593.98	858.90	889.76	780.72	808.02	966.14	1026.71	1,276.19	1087.01	8881.410
Chillies	804.792	794.12	774.87	760.98	839.71	811.14	751.61	780.37	684.47	702.05	7704.112
Coriander	557.870	543.20	447.13	552.66	673.97	581.60	532.32	470.16	528.91	656.46	5544.280
Garlic	242.491	247.52	230.59	262.06	320.93	280.95	316.98	357.96	352.14	392.15	3003.771
Turmeric	218.646	194.23	232.67	184.44	221.70	185.90	231.42	252.98	296.35	292.88	2311.216
Pepper	200.279	124.60	123.81	128.67	130.95	128.59	134.29	244.60	257.83	309.34	1782.959
Ginger	155.063	136.25	132.62	141.65	167.37	163.62	165.28	164.13	178.57	204.84	1609.393

Source: Department of Agriculture and Family Welfare

Table: 1.2**Titel: Yield of Top Produced Spices in India reference with the Fig. 1.3**

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Ginger	4.873	5.01	4.94	5.37	6.38	6.78	10.66	10.90	10.55	10.86	76.32
Garlic	5.065	5.09	5.43	5.44	5.28	5.76	5.08	8.13	8.36	8.13	61.77
Turmeric	5.337	5.00	5.11	4.50	4.76	5.07	3.73	3.80	3.89	3.84	45.04
Tamarind	3.467	3.47	3.21	3.70	4.03	3.68	4.16	4.14	4.01	3.75	37.62
Chillies	1.586	1.64	1.93	2.11	2.50	1.87	2.86	2.23	2.82	2.92	22.47
Cinnamon/Tejpat	1.710	1.71	1.82	1.84	1.92	1.84	1.88	2.24	2.49	2.50	19.95
Fennel	1.436	1.44	1.29	1.55	1.68	1.70	1.57	1.74	1.70	1.54	15.65

Source: Department of Agriculture and Family Welfare

Table: 1.3**Titel: Production of Least Produced Spices in India reference with the Fig. 1.4**

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Cardamom	15.816	18.43	21.28	24.36	24.07	27.66	28.24	22.57	20.90	33.95	237.276
Mint	0.00	0.00	0.00		0.00	0.00	32.74	34.05	43.23	45.80	155.820
Nutmeg	12.574	12.62	12.78	14.40	14.46	15.41	15.11	15.47	14.97	15.59	143.384
cinnamon/T	5.035	5.05	5.05	5.05	5.07	5.04	5.09	5.11	7.56	4.88	52.935
Clove	1.105	1.10	1.07	1.22	1.21	1.22	1.22	1.23	1.24	1.18	11.795
Saffron/Var	1.074	1.07	1.07	1.05	0.29	0.32	0.32	0.14	0.10	0.04	5.474

Source: Department of Agriculture and Family Welfare

Table: 1.4

Titel: Area under Least Produced Spices in India reference with the Fig. 1.5

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Mint	0.00	0.00	0.00	0.00	0.00	0.00	327.36	333.90	328.04	347.28	1336.580
Dill/Poppy	33.470	33.47	33.47	24.27	25.64	35.56	35.91	29.79	32.80	25.28	309.660
Nutmeg	17.485	17.49	18.90	21.12	21.22	22.81	23.38	23.84	23.58	24.43	214.255
Saffron/Va	7.095	7.10	7.25	6.47	4.41	4.48	4.56	3.81	3.77	3.77	52.715
Cinnamon	2.944	2.95	2.77	2.74	2.76	2.62	2.70	2.28	3.04	1.96	26.764
Clove	2.387	2.38	2.06	2.32	2.31	2.19	2.18	2.33	2.20	1.94	22.297

Source: Department of Agriculture and Family Welfare

Table: 1.5

Titel: Yield of Least Produced Spices in India reference with the Fig. 1.6

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Cumin	0.664	0.66	0.60	0.55	0.62	0.63	0.71	0.68	0.71	0.99	6.814
Clove	0.463	0.46	0.52	0.53	0.52	0.56	0.56	0.53	0.56	0.61	5.313
Pepper	0.203	0.42	0.41	0.50	0.42	0.55	0.49	0.56	0.41	0.45	4.413
Cardamom	0.178	0.20	0.23	0.24	0.28	0.33	0.34	0.28	0.24	0.40	2.718
Vanilla	0.151	0.15	0.15	0.16	0.07	0.07	0.070	0.39	0.40	0.40	2.011
Mint	0.000	0.00	0.00	0.00	0.00	0.00	0.10	0.10	0.13	0.13	0.460

Source: Department of Agriculture and Family Welfare

Table: 1.6

Titel: Production of Middel range Produced Spices in India reference with the Fig. 1.7

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Fenugreek	93.605	93.12	65.94	123.35	219.18	210.15	148.99	122.22	126.51	156.16	1359.225
Cardamom	89.006	92.39	92.84	99.56	85.66	84.68	84.20	81.11	85.42	85.01	879.876
Fennel	99.554	99.61	54.16	38.66	75.93	90.85	65.98	90.24	82.60	82.77	780.354
Tmarind	58.428	58.11	58.59	54.48	52.89	48.89	48.42	45.69	47.29	41.63	514.418
Dill/Poppy/C	35.376	35.38	26.67	24.09	23.71	30.83	34.76	34.91	41.13	42.01	328.866

Source: Department of Agriculture and Family Welfare

Table: 1.7**Titel: Area under Middel range Produced Spices in India reference with the Fig. 1.8**

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Mint	0.00	0.00	0.00	0.00	0.00	0.00	327.36	333.90	328.04	347.28	1336.580
Dill/Poppy/C	33.470	33.47	33.47	24.27	25.64	35.56	35.91	29.79	32.80	25.28	309.660
Nutmeg	17.485	17.49	18.90	21.12	21.22	22.81	23.38	23.84	23.58	24.43	214.255
Saffron/Vani	7.095	7.10	7.25	6.47	4.41	4.48	4.56	3.81	3.77	3.77	52.715
Cinnamon	2.944	2.95	2.77	2.74	2.76	2.62	2.70	2.28	3.04	1.96	26.764
Clove	2.387	2.38	2.06	2.32	2.31	2.19	2.18	2.33	2.20	1.94	22.297

Source: Department of Agriculture and Family Welfare

Table: 1.8**Titel: Yield of Middel range Produced Spices in India reference with the Fig. 1.9**

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Cumin	0.664	0.66	0.60	0.55	0.62	0.63	0.71	0.68	0.71	0.99	6.814
Clove	0.463	0.46	0.52	0.53	0.52	0.56	0.56	0.53	0.56	0.61	5.313
Pepper	0.203	0.42	0.41	0.50	0.42	0.55	0.49	0.56	0.41	0.45	4.413
Cardamom	0.178	0.20	0.23	0.24	0.28	0.33	0.34	0.28	0.24	0.40	2.718
Vanilla	0.151	0.15	0.15	0.16	0.07	0.07	0.070	0.39	0.40	0.40	2.011
Mint	0.000	0.00	0.00	0.00	0.00	0.00	0.10	0.10	0.13	0.13	0.460

Source: Department of Agriculture and Family Welfare

Table: 1.9

Area, Production and Yield of Flowers

Titel: Production of top Produced Flowers in India reference with the Fig. 2.1

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Rose	27467.25	19978.42	262.560	120.51	301.95	377.54	368.55	382.65	430.47	498.23	50188.13
Gladious	10616.88	7080.13	143.588	77.12	174.73	178.40	165.70	137.47	149.33	260.77	18984.12
Tube Rose	14051.94	1588.41	54.119	39.89	197.20	189.02	199.60	226.14	235.36	215.54	16997.22
Chrysanthemum	1671.78	206.19	185.090	106.25	200.16	315.71	479.26	536.35	601.51	471.85	4774.15

Source: Department of Agriculture and Family Welfare

Table: 2.1**Titel: Area under top Produced Flowers in India reference with the Fig. 2.2**

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Marigold	44.00	42.88	55.89	54.90	66.10	68.03	76.08	72.74	79.70	73.99	634.31
Rose	27.87	28.13	30.87	26.38	29.41	33.94	35.57	32.82	36.00	37.55	318.54
Jasmine	41.74	10.35	12.25	9.91	22.86	24.54	25.53	22.97	22.04	26.15	218.34
Chrysanthemum	18.68	18.36	16.63	10.70	20.09	20.80	25.76	22.09	25.28	24.02	202.41

Source: Department of Agriculture and Family Welfare

Table: 2.2**Titel: Yield of top Produced Flowers in India reference with the Fig. 2.3**

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Carnation	2.11	2.54	2.66	2.73	2.58	4.71	4.42	4.69	7.48	8.23	42.15
Gebera	18.11	18.29	19.32	20.16	22.22	24.77	29.23	38.76	40.02	41.23	272.11
Anthurium	3.20	4.15	1.90	3.71	8.63	7.32	10.14	13.96	2.83	0.90	56.74
Gladious	4.77	5.11	4.18	6.22	5.76	5.76	5.19	4.60	3.32	3.48	48.39

Source: Department of Agriculture and Family Welfare

Table: 2.3

Titel: Production of Least Produced Flowers in India reference with the Fig. 2.4

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Anthurium	120.09	320.27	2.973	0.43	3.22	3.35	4.44	9.69	9.00	2.91	476.37
Orchids	0.28	0.18	7.958	2.54	4.88	14.33	14.85	14.52	14.82	14.69	89.05
Tulip	40.00	45.10	0.25	0.00	0.10	0.10	0.10	0.21	0.08	0.08	86.02

Source: Department of Agriculture and Family Welfare

Table: 2.4**Titel: Area under Least Produced Flowers in India reference with the Fig. 2.5**

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Carnation	0.04	0.01	0.21	0.23	3.08	3.20	3.16	0.45	2.35	2.40	15.13
Orchids	0.00	0.01	0.43	0.47	0.82	0.79	0.80	0.85	2.91	2.94	10.02
Tulip	0.01	0.02	0.02	0.02	0.02	0.02	0.02	0.01	0.02	0.02	0.18

Source: Department of Agriculture and Family Welfare

Table: 2.5**Titel: Yield of Least Produced Flowers in India reference with the Fig. 2.6**

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Gladious	11.26	11.91	12.31	12.50	14.75	14.64	12.90	12.92	12.79	22.16	138.14
Orchids	9.46	14.75	18.522	5.46	5.94	18.05	18.49	17.09	5.09	4.99	117.84
Rose	8.50	8.74	9.860	10.11	10.27	11.12	10.36	11.66	11.96	13.27	105.85
Tube rose	5.26	5.74	6.97	6.00	12.18	11.68	10.39	11.13	10.67	9.81	89.83

Source: Department of Agriculture and Family Welfare

Table: 2.6

Titel: Production of Middel range Produced Flowers in India reference with the Fig. 2.7

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Marigold	381.98	360.21	515.561	485.16	608.89	640.51	811.09	829.92	887.16	780.11	6300.59
Jasmine	206.67	50.89	66.930	56.14	199.26	199.25	195.11	182.81	176.68	240.23	1573.97
Gerbera	319.82	249.20	21.804	4.12	25.55	41.57	38.62	50.37	43.96	26.16	821.17
Carnation	365.68	148.70	6.597	0.64	7.96	15.09	10.81	18.23	17.56	19.72	610.99

Source: Department of Agriculture and Family Welfare

Table: 2.7

Titel: Area under Middel range Produced Flowers in India reference with the Fig. 2.8

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Tube Rose	12.45	7.95	7.77	6.65	16.19	16.18	19.22	20.32	22.07	21.97	150.77
Gladious	11.66	9.37	11.67	11.16	11.85	12.18	12.85	10.64	11.68	11.77	114.83
Gerbera	0.11	0.06	0.82	0.90	1.15	1.68	1.32	1.30	5.48	5.37	18.19
Anthurium	0.10	0.07	0.13	0.12	0.37	0.46	0.44	0.26	3.18	3.22	8.35

Source: Department of Agriculture and Family Welfare

Table: 2.8

Titel: Yield of Middel range Produced Flowers in India reference with the Fig. 2.9

	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	Total
Crops											
Chrysanthemum	12.50	11.23	11.13	9.93	9.96	15.18	18.61	24.28	23.80	19.65	156.27
Marigold	8.68	8.40	9.224	8.84	9.21	9.41	10.66	11.41	11.13	10.54	97.50
Jasmine	4.95	4.92	5.463	5.67	8.72	8.12	7.64	7.96	8.02	9.19	70.65

Source: Department of Agriculture and Family Welfare

Table: 2.9